

● RS002 Week 15 — English Worksheet

Topic: Multiple Files — `import` Your Own Data · **Course:** Pokédex App · **Time:** about 45 minutes

This week your project grows into **two files**. The Pokémon data moves into `pokemon_data.py`, and your game lives in `main.py`, which pulls the data in with `from pokemon_data import pokedex`. Splitting files is how real programs stay organised.

1 · Recall

From memory:

Write a function `battle(p1, p2)` **that returns whichever Pokémon has the higher** `hp + attack` **(return** `None` **on a tie).**

2 · Predict 🧠

Read each block. Before you run it, write what you think will happen.

Two files:

```
# pokemon_data.py
pokedex = [
    {"name": "피카츄", "type": "전기"},
    {"name": "파이리", "type": "불"},
]
```

```
# main.py
from pokemon_data import pokedex
print(len(pokedex))
print(pokedex[0]["name"])
```

When you run `main.py`, what two lines print?


```
# main.py
import random
from pokemon_data import pokedex

print(random.choice(pokedex)["name"])
```

Both `import` styles appear here. What does this line print? Could it change each run?


```
# main.py
from pokemon_data import pokedex
print(pokemon_data)
```

`from ... import pokedex` only brings in `pokedex`. What happens on the `print` line?

3 · Spot the Bug 🐛

Read what each block is **supposed** to do, fix it, then explain the fix.

Bug A — `main.py` should pull in the dex from the data file. Right now nothing is imported, so `pokedex` does not exist.

```
# main.py
print(len(pokedex))
```

Hint: add a `from pokemon_data import pokedex` line at the top.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The data lives in `pokemon_data.py`, but this import names the wrong file.

```
# main.py
from pokemon import pokedex
print(pokedex[0]["name"])
```

Hint: the module name must match the file name (without `.py`).

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — `random.choice` is used but only the data is imported. The program crashes on `random`.

```
# main.py
from pokemon_data import pokedex
print(random.choice(pokedex)["name"])
```

Hint: `random` is a separate module — import it too.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Start from this working two-file setup:

```
# pokemon_data.py – data only
pokedex = [
    {"name": "피카츄", "type": "전기", "hp": 35, "attack": 55},
    {"name": "이상해씨", "type": "풀", "hp": 45, "attack": 49},
    {"name": "파이리", "type": "불", "hp": 39, "attack": 52},
    {"name": "꼬부기", "type": "물", "hp": 44, "attack": 48},
]
```

```

# main.py - the game
import random
from pokemon_data import pokedex

def show_pokemon(p):
    print(f"{p['name']} ({p['type']}) - HP {p['hp']}, ATK {p['attack']}")

def battle(p1, p2):
    score1 = p1["hp"] + p1["attack"]
    score2 = p2["hp"] + p2["attack"]
    if score1 > score2:
        return p1
    elif score2 > score1:
        return p2
    return None

trainer = input("트레이너 이름: ").strip() or "이름 모를 트레이너"
print(f"\n환영합니다, {trainer} 트레이너님!")

print("\n=== 포켓몬 도감 ===")
for i, p in enumerate(pokedex):
    print(f"{i+1}. {p['name']}")

choice = input("\n원하는 포켓몬 이름: ").strip()
my_pokemon = None
for p in pokedex:
    if p["name"] == choice:
        my_pokemon = p
        break

if my_pokemon == None:
    print("도감에 없는 포켓몬이어서 피카츄로 시작합니다.")
    my_pokemon = pokedex[0]

enemy = random.choice(pokedex)
print(f"\n🔴 배틀 시작!")
show_pokemon(my_pokemon)
print("    vs")
show_pokemon(enemy)

winner = battle(my_pokemon, enemy)
if winner == my_pokemon:
    print(f"\n🏆 {trainer} 트레이너님의 승리!")
elif winner == enemy:
    print(f"\n😞 졌습니다... 다시 도전해보세요!")

```

```
else:  
    print(f"\n🏆 무승부!")
```

Make these changes one at a time and run after each:

1. Add two more Pokémon in `pokemon_data.py` **only** and confirm they appear in `main.py`.
2. Move your `show_pokemon` style (emojis, layout) to your taste.
3. Add a second list `wild_pokemon` to the data file and `import` it as well.

Write your changed / added lines here:

5 • Make It

Split your Week 14 app into **two files**: `pokemon_data.py` for the dex, `main.py` for everything else. `main.py` imports the dex and still runs the full app.

When it works, send a **video or screenshots of both files** on KakaoTalk. Then explain what you did. Use these starters — write 4 to 6 sentences.

First, I moved ... into `pokemon_data.py` .

In `main.py` I brought it in with ...

```
from pokemon_data import pokedex
```

 means ...

`import X` and `from X import Y` are different because ...

The trickiest part was ...

Splitting files is helpful because ...

6 · Record Your Walkthrough 🎥

Film both files and the app running. Try to use: **import, module, from import, data file, main file.**

1. Show `pokemon_data.py` and read its `pokedex` .
2. Show the `import` line in `main.py` and explain it.
3. Run `main.py` and play through one battle.
4. Add one Pokémon to the data file live and show it appear.

Plan what you will say here first:

Submit ✓

Send this worksheet + **both files** + your walkthrough video to teacher on KakaoTalk.